EXPERIMENT : USER MANAGEMENT AND ROLE-BASED ACCESS CONTROL (RBAC)

AIM

To implement, configure, and evaluate a comprehensive User Management and Role-Based Access Control (RBAC) architecture within a low-code application using Zoho Creator. The objective is to define multi-tier hierarchical organizational roles, enforce granular permission profiles across system modules, establish record-level data sharing rules, and map administrative users to ensure data confidentiality, integrity, and controlled operational governance.

APPLICATION CONTEXT & SYSTEM OVERVIEW

The target system configured for this experiment is the 'Vehicle Service & Repair Request Management' enterprise application. In this operational workflow, multi-tier access control is mandatory to prevent unauthorized data manipulation, protect customer repair histories, and establish strict segregation of duties between management and maintenance personnel.

PROCEDURE

Step 1: Navigating to Application Governance Settings

Access the Zoho Creator application development workbench. Open the 'Vehicle Service & Repair Request Management' system in edit mode, select the 'Settings' gear icon located in the upper application navigation bar, and select 'User Permissions' under the Permissions administration panel to access the user control console.

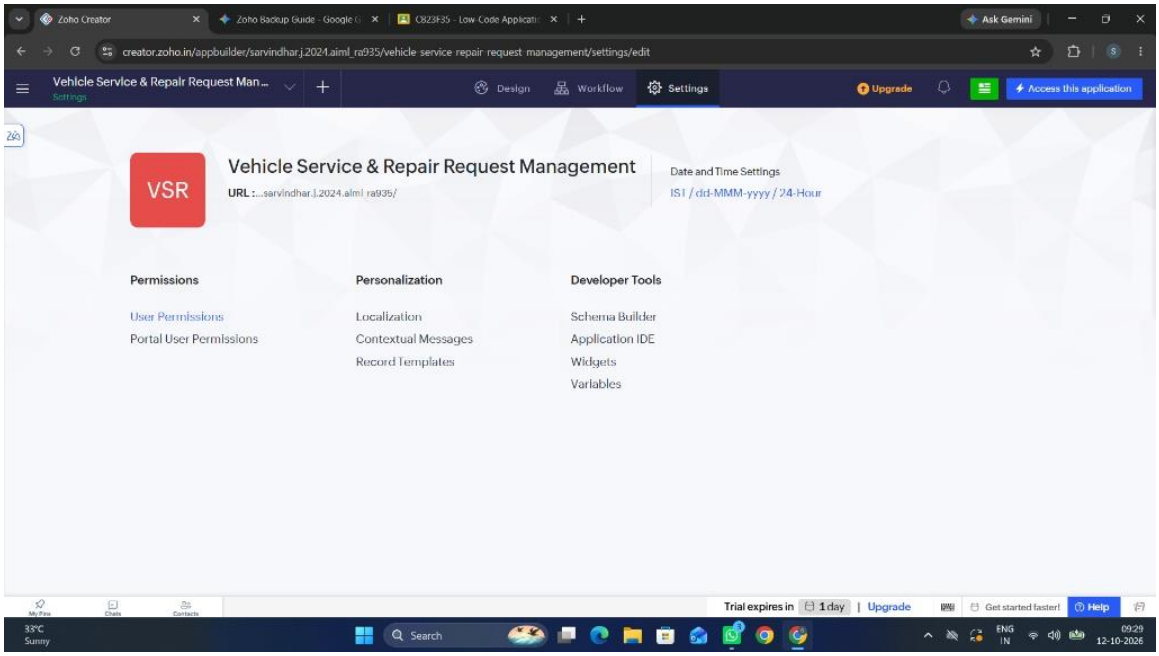


Figure 1: Application Settings & User Permissions Configuration Panel

Step 2: Inspecting Permission Profile Repository

Review the baseline permission sets available within the application under the 'Permissions' repository tab. The environment provides default administrative and standard profiles ('Read' and 'Write'). Initiate the creation of customized, role-tailored security profiles by choosing '+ Add Permission'.

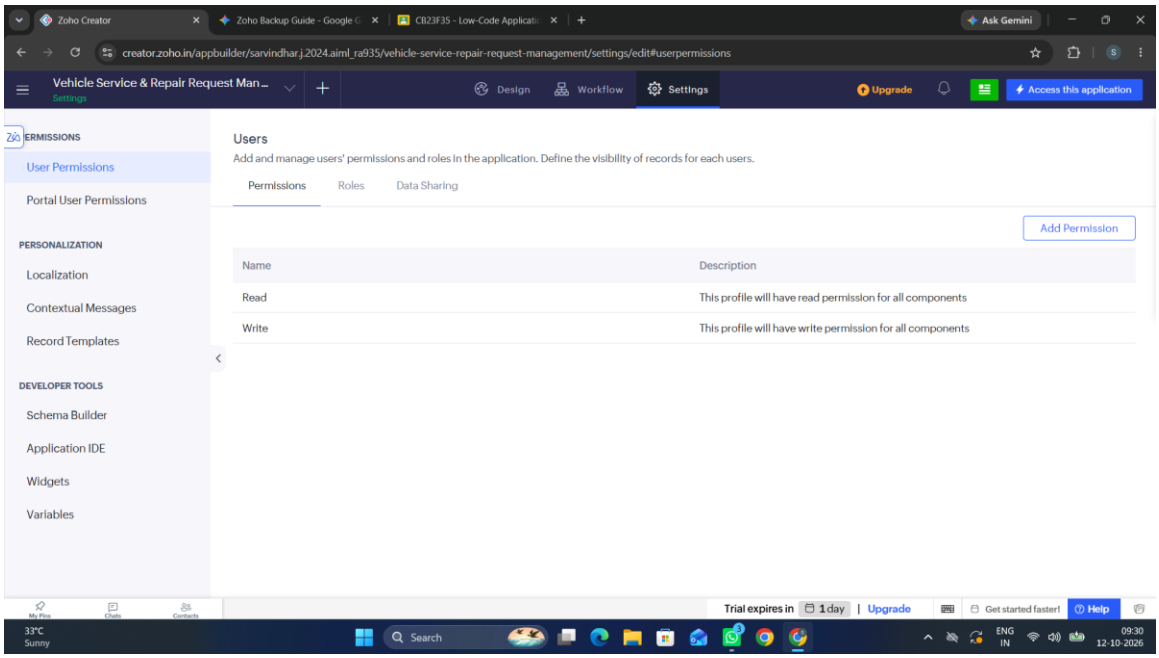


Figure 2: Users Permission Sets Repository Interface

Step 3: Configuring the Manager Permission Set (Full Control)

Construct an elevated permission profile named 'Manager Access'. Within the module privilege matrix, assign comprehensive permissions—encompassing Access, View, Edit, and Delete privileges—across all critical operational modules including Service_Request_Form, Vehicle_Master, and Service_Catalog to empower managerial governance.

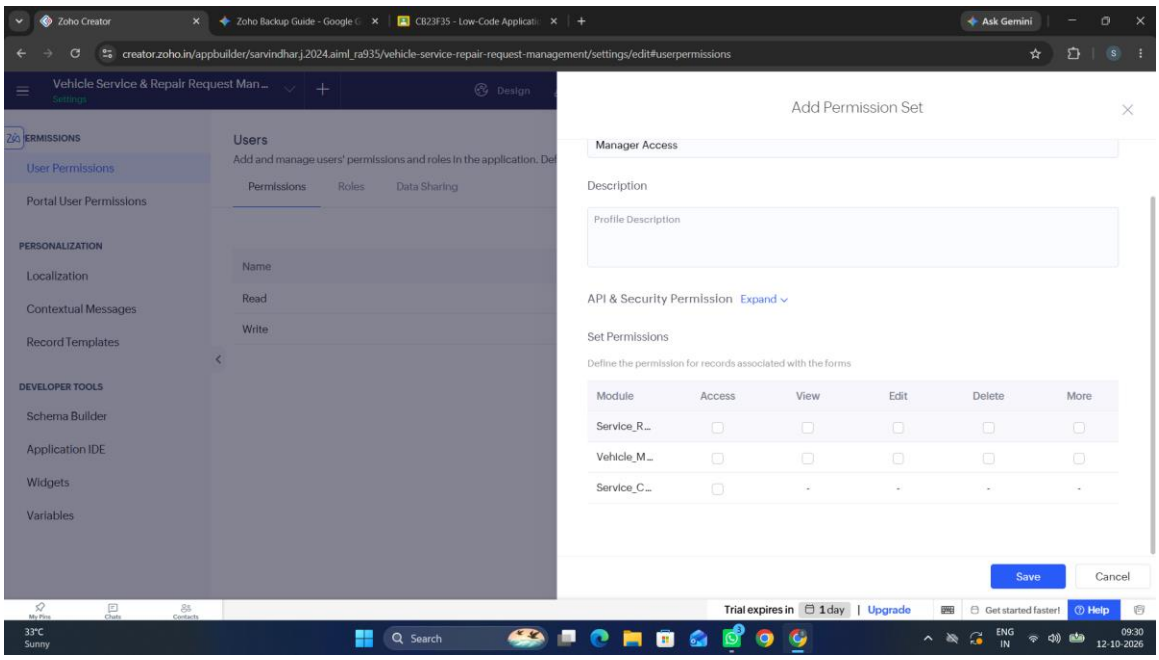


Figure 3: Configuration of 'Manager Access' Permission Set with Full Privileges

Step 4: Configuring the Mechanic Permission Set (Restricted Access)

Construct a restricted operational profile titled 'Mechanic Access'. Grant View and Edit privileges across relevant service order forms while explicitly revoking Delete access. This enforces the Principle of Least Privilege (PoLP), ensuring field mechanics can view and update work orders without the capability to delete service records.

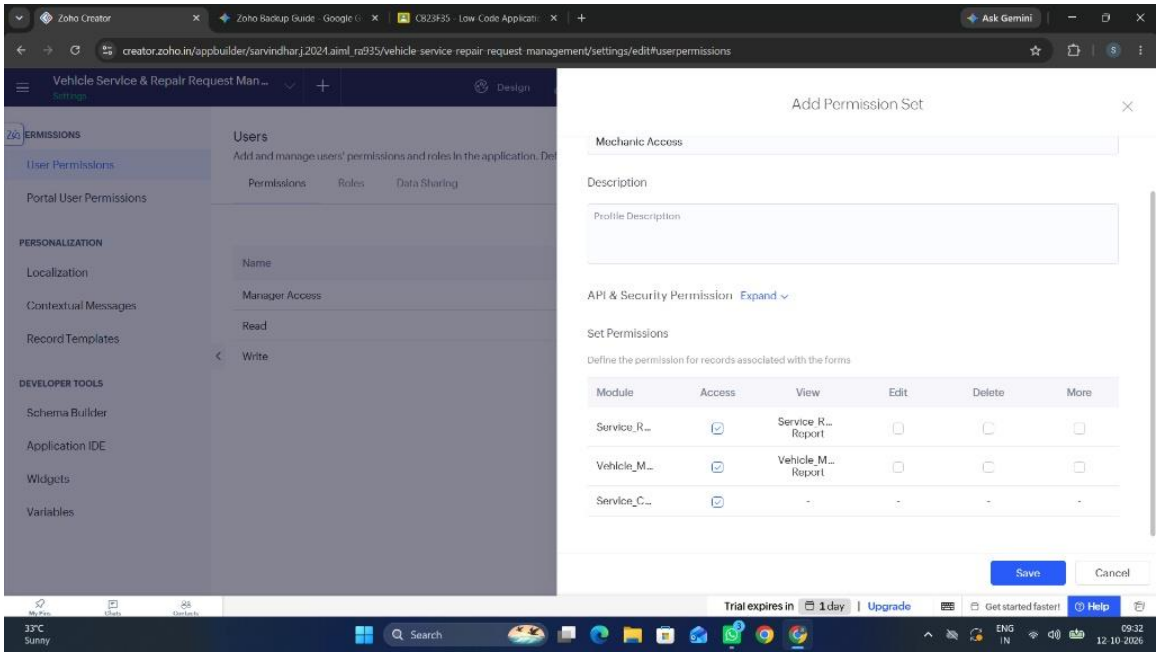


Figure 4: Configuration of 'Mechanic Access' Permission Set with Revoked Delete Privileges

Step 5: Establishing Hierarchical Role Structure

Navigate to the 'Roles' tab to establish organizational reporting hierarchy. Under the existing root node ('CEO'), instantiate a managerial role titled 'Service Manager'. Subsequently, define the operational subordinate role titled 'mechanic', explicitly setting its reporting parent to 'Service Manager' to form a structured reporting tree.

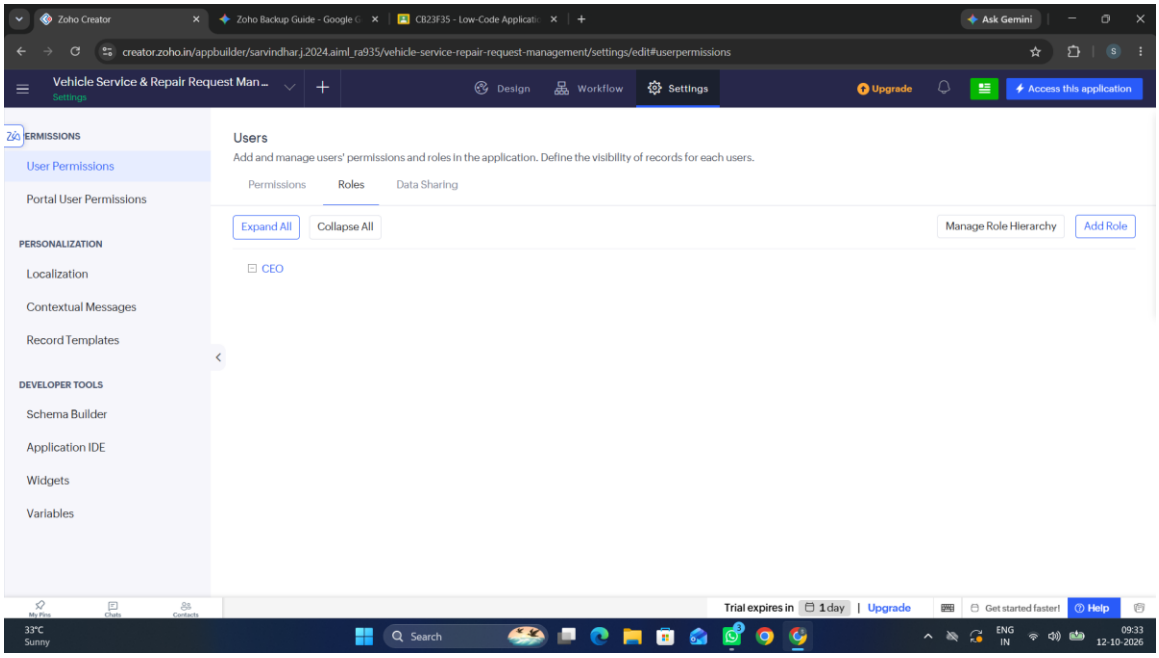


Figure 5: Organizational Role Hierarchy Definition

Step 6: Defining Record-Level Data Sharing Rules

Navigate to the 'Data Sharing' configuration tab and establish a custom record sharing rule named 'Share Mechanic Requests with Manager'. Configure the rule parameters such that records instantiated from the 'mechanic' role in 'Service_Request_Form' are dynamically shared to the 'Service Manager' role with 'Read/Write' access, guaranteeing complete managerial oversight.

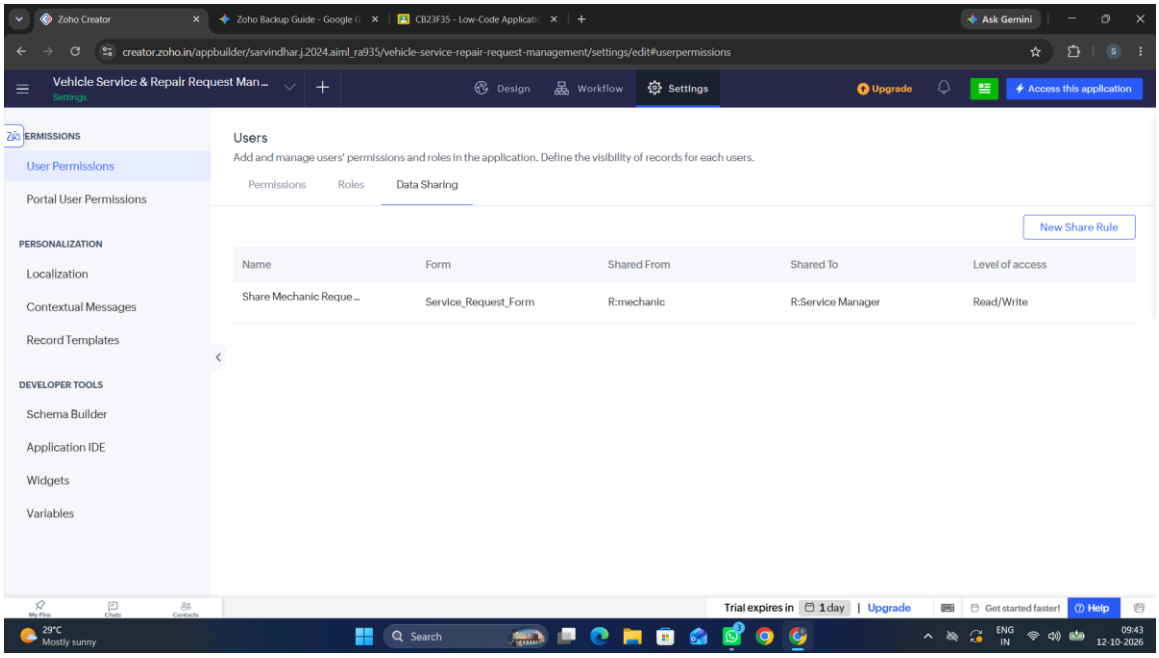


Figure 6: Custom Data Sharing Rule Mapping from Mechanic to Service Manager

Step 7: User Provisioning and RBAC Assignment

Switch to the Zoho Creator administrative console under Manage > Users. Execute '+ Add User' to provision a new user account ('Mechanic', jsarvindhar@gmail.com). Bind the provisioned user specifically to the 'Vehicle Service & Repair Request Management' application, assigning the 'mechanic' Role and 'Mechanic Access' Permission Set.

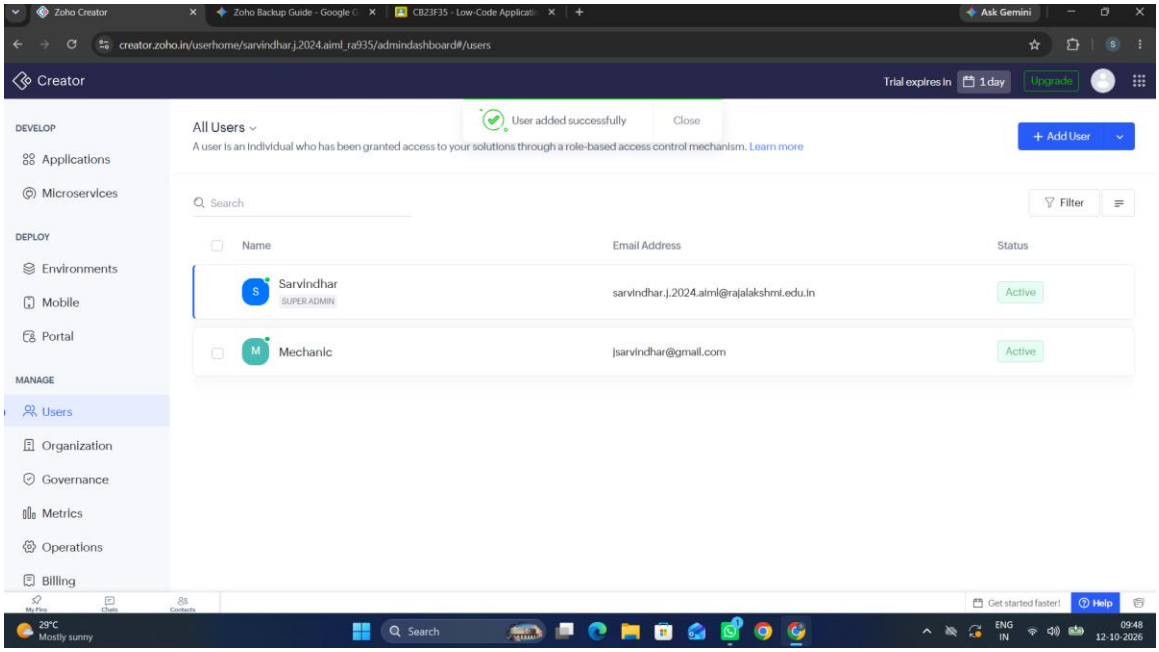


Figure 7: Successfully Provisioned User Repository with Mapped Roles and Active Status

RESULT

Thus, the User Management and Role-Based Access Control (RBAC) framework was successfully implemented and validated within the Zoho Creator application. Multi-tier organizational role hierarchies ('CEO' -> 'Service Manager' -> 'mechanic'), granular permission profiles ('Manager Access' and 'Mechanic Access'), data sharing rules, and user assignment were effectively configured and verified, fulfilling the principles of least privilege and strict data governance.